Introduction to Lie Group Theory

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Chapter 1

Matrix Groups

1.1 Groups of Matrices

Let $M_{m,n}(\mathbb{K})$ be the set of $m \times n$ matrices whose entries are in $\mathbb{K}$, a commutative field. We will also use the special notations $M_n(\mathbb{K}) = M_{n.n}(\mathbb{K})$, and $\mathbb{K}^n = M_{n,1}(\mathbb{K})$.

$M_{m,n}(\mathbb{K})$ is a vector space over $\mathbb{K}$ with dimension mn. As well as being a vector space over $\mathbb{K}$ with dimension n^2, $M_n(\mathbb{K})$ is also a ring with the usual addition and multiplication of matrices. It is also an important example of a finite dimensinoal algebra over $\mathbb{K}$.

We will use the notation

$$\mathrm{GL}_n(\mathbb{K}) = \{A \in M_n(\mathbb{K}) : \det A \neq 0\}$$

for the set of invertible $n \times n$ matrices, known as the $n \times n$ *general linear group*, and

$$\mathrm{SL}_n(\mathbb{K}) = \{A \in M_n(\mathbb{K}) : \det A = 1\} \subseteq \mathrm{GL}_n(\mathbb{K})$$

for the set of $n \times n$ unimodular matrices, known as the $n \times n$ *special linear group*.

Theorem 1.1. *The sets $\mathrm{GL}_n(\mathbb{K})$ and $\mathrm{SL}_n(\mathbb{K})$ are groups under matrix multiplication. Furthermore, $\mathrm{SL}_n(\mathbb{K})$ is a subgroup of $\mathrm{GL}_n(\mathbb{K})$.*

Since determinant of a matrix is a polynomial in the entries of the matrix, it can be proved that the determinant function

$$\det : M_n(\mathbb{K}) \longrightarrow \mathbb{K}$$

is continous.

Proposition 1.1. *Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Then*

i) $GL_n(\mathbb{K}) \subseteq M_n(\mathbb{K})$ *is an open subset;*

ii) $SL_n(\mathbb{K}) \subseteq M_n(\mathbb{K})$ *is a closed subset.*

Proof. $GL_n(\mathbb{K}) = M_n(\mathbb{K}) - \det^{-1}\{0\}$, which is open since $\{0\}$ is closed. Similarly, $SL_n(\mathbb{K}) = \det^{-1}\{1\}$, which is closed since the singleton $\{1\}$ is closed in $\mathbb{K}$. ■

Definition 1.1. Let G be a topological space, and view $G \times G$ as the product space. Suppose that G is also a group with multiplication map mult : $G \times G \longrightarrow G$, and inverse map inv : $G \longrightarrow G$. G is a *topological group* if mult and inv are continous.

Theorem 1.2. *For $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, $GL_n(\mathbb{K})$ and $SL_n(\mathbb{K})$ are topological groups.*

1.2 Matrix Groups

Definition 1.2. A subgroup G of $GL_n(\mathbb{K})$ is called a *matrix group* if it is a closed subset of $GL_n(\mathbb{K})$.

Remark. We may consider $GL_n(\mathbb{K})$ as a subgroup of $GL_{n+1}(\mathbb{K})$ by identifying the $n \times n$ matrix $A = [a_{ij}]$ with

$$\begin{bmatrix} A & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & \cdots & a_{1n} & 0 \\ \vdots & \ddots & \vdots & \vdots \\ a_{n1} & \cdots & a_{nn} & 0 \\ 0 & \cdots & 0 & 1 \end{bmatrix}$$

And it is easily verified that $GL_n(\mathbb{K})$ is closed in $GL_{n+1}(\mathbb{K})$. With the aid of this embedding, any matrix subgroup of $GL_n(\mathbb{K})$ can also be viewed as a matrix subgroup of $GL_{n+1}(\mathbb{K})$.

Definition 1.3. Let G, H be two matrix groups. A groups homomorphism $\varphi : G \longrightarrow H$ is a *continuous homomorphism of matrix groups* if it is continuous and its image $\varphi(G)$ is a closed subgroup of H.

Examples. Some important examples of real and complex matrix groups:

1. **Groups of Upper Triangular Matrices**

   For $n \geqslant 1$, an $n \times n$ matrix $A = [a_{ij}]$ is said to be *upper triangular* if $a_{ij} = 0$ if $i < j$. An upper traingular matrix is said to be *unipotent* of all of it's diagonal entries are 1.

The *upper triangular subgroup* or *Borel subgroup* of $\mathrm{GL}_n(\mathbb{K})$ is

$$\mathrm{UT}_n(\mathbb{K}) = \{A \in \mathrm{GL}_n(\mathbb{K}) : \text{A is upper triangular}\},$$

while the *unipotent* subgroup of $\mathrm{GL}_n(\mathbb{K})$ is

$$\mathrm{SUT}_n(\mathbb{K}) = \{A \in \mathrm{GL}_n(\mathbb{K}) : \text{A is unipotent}\}.$$

It is easy to verify that $\mathrm{UT}_n(\mathbb{K})$ and $\mathrm{SUT}_n(\mathbb{K})$ are closed subgroups of $\mathrm{GL}_n(\mathbb{K})$, and that $\mathrm{SUT}_n(\mathbb{K})$ is a subgroup of $\mathrm{UT}_n(\mathbb{K})$.

2. **Affine Groups**

   The *n-dimensional affine group* over $\mathbb{K}$ is

$$\mathrm{Aff}_n(\mathbb{K}) = \left\{ \begin{bmatrix} A & t \\ 0 & 1 \end{bmatrix} : A \in \mathrm{GL}_n(\mathbb{K}), t \in \mathbb{K}^n \right\}$$

   which is clearly a closed subgroup of $\mathrm{GL}_{n+1}(\mathbb{K})$. If we identify $x \in \mathbb{K}^n$ with $\begin{bmatrix} x \\ 1 \end{bmatrix} \in \mathbb{K}^{n+1}$, then as a consequence of

$$\begin{bmatrix} A & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix} = \begin{bmatrix} Ax + t \\ 1 \end{bmatrix}$$

   we obtain an action of $\mathrm{Aff}_n(\mathbb{K})$ on $\mathbb{K}^n$.

   The group $\mathrm{Aff}_n(\mathbb{K})$ can be expressed as the semi-direct product of $\mathrm{Trans}_n(\mathbb{K})$ and $\mathrm{GL}_n(\mathbb{K})$, where

$$\mathrm{Trans}_n(\mathbb{K}) = \left\{ \begin{bmatrix} I_n & t \\ 0 & 1 \end{bmatrix} : t \in \mathbb{K}^n \right\}$$

   is a closed normal subgroup of $\mathrm{Aff}_n(\mathbb{K})$. That is,

$$\mathrm{Aff}_n(\mathbb{K}) = \mathrm{GL}_n(\mathbb{K}) \ltimes \mathrm{Trans}_n(\mathbb{K}).$$

3. **Orthagonal Groups**

4. **Unitary Groups**

5. **Symplectic Groups**

1.3 Complex Matrices as Real Matrices

1.4 Matrix Groups of Normed Vector Spaces

1.5 Matrix Exponential and Logarithm

1.6 One Parameter Subgroups in Matrix Groups

Chapter 2

Tangent Spaces and Lie Algebras

2.1 Lie Algebras

2.2 Curves and Tangent Spaces

2.3 Lie Algebras of Matrix Groups

2.4 $SO(3)$ and $SU(2)$

2.5 Complexification of Real Lie Algebra

Chapter 3

Lie Groups

3.1 Smooth Manifolds

3.2 Tangent Spaces and Derivatives

3.3 Lie Groups

3.4 Matrix Groups are Lie Groups

3.5 Not All Lie Groups are Matrix Groups

References

[Baker, 2002] Baker, A. (2002). *Matrix Groups*. Springer Undergraduate Mathematics Series. Springer-Verlag.

[Trapp, 2005] Trapp, K. (2005). *Matrix Groups for Undergraduates*. Student Mathematical Library. American Mathematical Society, second edition.